

Supporting information – 3D Imaging Using Extreme Dispersion in Optical Metasurfaces

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Reconstruction network architectures

The RGB reconstruction and depth prediction networks share similar encoder-decoder (with the same number of layers) architectures, except for the output layer. The detailed architectures are shown in Tables S1 and S2. For RGB reconstruction, the network learns a 3-channel residual image (with a tanh activation) and outputs the sum of the residual image and the input. For depth estimation, the network directly outputs a single-channel depth map (with a sigmoid activation). The encoder consists of convolutional blocks (two 3×3 convolutions, plus a rectified linear unit (ReLU) and a batch normalization (BN)), followed by 2×2 max pooling downsamplings. The decoder is composed of 2×2 upsampling layers and 3×3 convolutional blocks. These hyperparameters are selected based on the previous works (1,2). They have been shown to have good performance in all-in-focus image and depth reconstruction.

Table S1: Detailed network architecture for RGB image reconstruction.

Name	Layer description	Output dimension	Input
down1_1	3×3 , 32 conv	$H \times W \times 32$	input image
down1_2	3×3 , 32 conv	$H \times W \times 32$	down1_1
down2_0	2×2 , max_pool	$1/2H \times 1/2W \times 32$	down1_2
down2_1	3×3 , 64 conv	$1/2H \times 1/2W \times 64$	down2_0
down2_2	3×3 , 64 conv	$1/2H \times 1/2W \times 64$	down2_1
down3_0	2×2 , max_pool	$1/4H \times 1/4W \times 64$	down2_2
down3_1	3×3 , 128 conv	$1/4H \times 1/4W \times 128$	down3_0
down3_2	3×3 , 128 conv	$1/4H \times 1/4W \times 128$	down3_1
down4_0	2×2 , max_pool	$1/8H \times 1/8W \times 128$	down3_2
down4_1	3×3 , 256 conv	$1/8H \times 1/8W \times 256$	down4_0
down4_2	3×3 , 256 conv	$1/8H \times 1/8W \times 256$	down4_1
down5_0	2×2 , max_pool	$1/16H \times 1/16W \times 256$	down4_2
down5_1	3×3 , 512 conv	$1/16H \times 1/16W \times 512$	down5_0
down5_2	3×3 , 512 conv	$1/16H \times 1/16W \times 512$	down5_1
up4_0	2×2 , upsamp	$1/8H \times 1/8W \times 256$	down5_2
up4_1	3×3 , 256 conv	$1/8H \times 1/8W \times 256$	[up4_0, down4_2]
up4_2	3×3 , 256 conv	$1/8H \times 1/8W \times 256$	up4_1
up3_0	2×2 , upsamp	$1/4H \times 1/4W \times 128$	up4_2
up3_1	3×3 , 128 conv	$1/4H \times 1/4W \times 128$	[up3_0, down3_2]
up3_2	3×3 , 128 conv	$1/4H \times 1/4W \times 128$	up3_1
up2_0	2×2 , upsamp	$1/2H \times 1/2W \times 64$	up3_2
up2_1	3×3 , 64 conv	$1/2H \times 1/2W \times 64$	[up2_0, down2_2]
up2_2	3×3 , 64 conv	$1/2H \times 1/2W \times 64$	up2_1
up1_0	2×2 , upsamp	$H \times W \times 32$	up2_2
up1_1	3×3 , 32 conv	$H \times W \times 32$	[up1_0, down1_2]
up1_2	3×3 , 32 conv	$H \times W \times 32$	up1_1
up1_3	1×1 , 3 conv	$H \times W \times 3$	up1_2
output	$\tanh(\text{up1}_3) + \text{input}$	$H \times W \times 3$	up1_3

Table S2: Detailed network architecture for depth estimation.

Name	Layer description	Output dimension	Input
down1_1	3×3, 32 conv	H×W×32	input image
down1_2	3×3, 32 conv	H×W×32	down1_1
down2_0	2×2, max_pool	1/2H×1/2W×32	down1_2
down2_1	3×3, 64 conv	1/2H×1/2W×64	down2_0
down2_2	3×3, 64 conv	1/2H×1/2W×64	down2_1
down3_0	2×2, max_pool	1/4H×1/4W×64	down2_2
down3_1	3×3, 128 conv	1/4H×1/4W×128	down3_0
down3_2	3×3, 128 conv	1/4H×1/4W×128	down3_1
down4_0	2×2, max_pool	1/8H×1/8W×128	down3_2
down4_1	3×3, 256 conv	1/8H×1/8W×256	down4_0
down4_2	3×3, 256 conv	1/8H×1/8W×256	down4_1
down5_0	2×2, max_pool	1/16H×1/16W×256	down4_2
down5_1	3×3, 512 conv	1/16H×1/16W×512	down5_0
down5_2	3×3, 512 conv	1/16H×1/16W×512	down5_1
up4_0	2×2, upsamp	1/8H×1/8W×256	down5_2
up4_1	3×3, 256 conv	1/8H×1/8W×256	[up4_0,down4_2]
up4_2	3×3, 256 conv	1/8H×1/8W×256	up4_1
up3_0	2×2, upsamp	1/4H×1/4W×128	up4_2
up3_1	3×3, 128 conv	1/4H×1/4W×128	[up3_0,down3_2]
up3_2	3×3, 128 conv	1/4H×1/4W×128	up3_1
up2_0	2×2, upsamp	1/2H×1/2W×64	up3_2
up2_1	3×3, 64 conv	1/2H×1/2W×64	[up2_0,down2_2]
up2_2	3×3, 64 conv	1/2H×1/2W×64	up2_1
up1_0	2×2, upsamp	H×W×32	up2_2
up1_1	3×3, 32 conv	H×W×32	[up1_0,down1_2]
up1_2	3×3, 32 conv	H×W×32	up1_1
up1_3	1×1, 1 conv	H×W×1	up1_2
output	sigmoid(up1_3)	H×W×1	up1_3

References

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